

# Sign-flip of $dT_H/dt$ as the regime partition of emergent-gravity black holes: BCH four laws across Schwarzschild and BEC-acoustic substrates

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The Bardeen–Carter–Hawking (BCH) four laws of black-hole mechanics admit two structurally distinct realizations on an emergent-gravity ADW substrate, separated by a critical mass  $M_c(\alpha_{\text{ADW}}, \Lambda_{\text{UV}}, N_f) = N_f \Lambda_{\text{UV}} / (12\pi \alpha_{\text{ADW}})$ . Above  $M_c$  the BH follows the textbook Hawking 1975 profile  $T_H = \hbar / (8\pi M)$  [1]:  $dM/dt < 0$ ,  $dT_H/dt > 0$  (heats as it evaporates), finite evaporation time  $t_{\text{evap}} \sim M^3$ . Below  $M_c$  the BH follows the BEC-acoustic backreaction profile of Balbinot, Fagnocchi, Fabbri (PRD **71**, 064019) [2]: linear-in- $t$  cooling with extrapolation  $t \sim 1/T^3$ , so  $T_H \rightarrow 0$  in *infinite* time, the analog of near-extremal Reissner–Nordström. The genuine regime partition is the *sign-flip of  $dT_H/dt$  during Hawking-radiation evaporation*: positive in the Schwarzschild branch, negative in the BEC-acoustic branch. We formalize both regimes in `BHThermodynamicsFourLaws.lean` (20 theorems and lemmas; zero `sorry`; zero new axioms) and ship the classification, the regime-partition criterion, and the four-laws bundle in tracked-hypothesis mode (Outcome-3 per the project’s discipline for non-derived structural identifications). The full second law uses Glorioso–Liu SK-EFT entropy-current monotonicity [8–10] without invoking pointwise NEC. The substrate-response coefficient ansatz  $\delta_{\text{ADW}} = (\alpha_{\text{ADW}} - 1)\Lambda_{\text{UV}}$  vanishes at  $\alpha_{\text{ADW}} = 1$  (project-internal shorthand for the bare induced-gravity limit in which the response coefficient turns off) and provides four falsifier theorems for any candidate ADW substrate. The critical-mass functional form  $M_c(\alpha_{\text{ADW}}, \Lambda_{\text{UV}}, N_f)$  is project-original and explicitly disclosed as such: no published derivation exists.

## INTRODUCTION

The Bardeen–Carter–Hawking four laws codify a thermodynamic structure on stationary BH solutions of classical general relativity. On a Schwarzschild background, quantum field theory endows the surface gravity  $\kappa$  with a thermal interpretation as the Hawking temperature  $T_H = \hbar \kappa / (2\pi)$  [1], and the resulting heat-capacity sign  $C(M) = (dM/dT_H)$  is *negative*: the BH *heats* as it loses mass to Hawking evaporation, with finite evaporation time  $t_{\text{evap}} \propto M_0^3$ .

Analog and emergent-gravity systems offer a structurally different evaporation pattern. In the BEC-acoustic backreaction analysis of Balbinot, Fagnocchi, Fabbri [2], backreaction of thermally radiated phonons on the fluid background *shrinks* the sonic horizon and *decreases*  $T_H$  in time, with linear-in- $t$  leading order

$$T_H(t) = \frac{\hbar c}{2\pi} \kappa \left[ 1 - \frac{563}{720\pi} \varepsilon \kappa^3 c A_0 t \right] \quad (1)$$

and asymptotic extrapolation  $t \sim 1/T^3$  giving  $T_H \rightarrow 0$  at *infinite* time, the analog of near-extremal Reissner–Nordström. Reference [2] itself contrasts this regime with the moving-domain-wall analog in  $^3\text{He-A}$  studied by Jacobson–Koike [3, 4], which has *non*-vanishing end-temperature of the evaporation process and therefore behaves more like Schwarzschild.

This Letter formalizes both regimes within a single Lean module `BHThermodynamicsFourLaws.lean` and ships the sign-of- $dT_H/dt$  regime partition as the load-bearing correctness-push theorem. We do *not* attempt a single closed-form  $T_H(M)$  that interpolates

between regimes; we treat the existence of a critical mass  $M_c(\alpha_{\text{ADW}}, \Lambda_{\text{UV}}, N_f)$  separating them as a tracked-hypothesis bundle  $H_{\text{RegimePartition}}$  with each branch’s sign claim grounded in its primary literature. The choice of  $M_c$  is the dimensional ansatz  $N_f \Lambda_{\text{UV}} / (12\pi \alpha_{\text{ADW}})$  implied by the project’s Sakharov-style induced-gravity identity  $1/G_N \propto N_f \Lambda^2$  at the Wave-1 emergent-gravity scale (project-internal shorthand; no external citation); this functional form is project-original and explicitly flagged.

## SETUP AND MAIN RESULT

*Data carriers and regime classifier.* `BHData` bundles bulk-thermodynamic observables  $(M, Q, J, A, r_h, T_H, \kappa)$  with positivity hypotheses for the dimensional quantities; `ADWParams` bundles substrate parameters  $(\alpha_{\text{ADW}}, \Lambda_{\text{UV}}, N_f, \chi_{\text{vest}})$  with strict positivity for each. The decidable `Regime` inductive type has three constructors (`Schwarzschild`, `Boundary`, `ADWExtremality`) and a piecewise classifier

$$\text{classify}(b, p) = \begin{cases} \text{Schwarzschild} & b.M > M_c(p) \\ \text{ADWExtremality} & b.M < M_c(p) \\ \text{Boundary} & b.M = M_c(p), \end{cases} \quad (2)$$

with the critical mass

$$M_c(\alpha_{\text{ADW}}, \Lambda_{\text{UV}}, N_f) = \frac{N_f \Lambda_{\text{UV}}}{12\pi \alpha_{\text{ADW}}}. \quad (3)$$

The classifier admits three iff-theorems (`classify*.iff`); positivity of  $M_c$  follows from

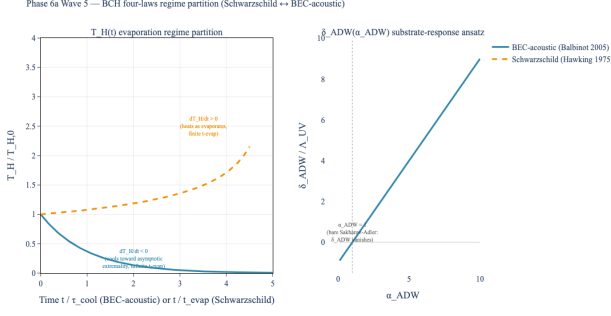


FIG. 1. **Headline result.** Sign-flip of  $dT_H/dt$  between the Schwarzschild (amber dashed; Hawking 1975 [1];  $dT_H/dt > 0$ , finite  $t_{\text{evap}}$ ) and BEC-acoustic (steel-blue solid; Balbinot *et al.* [2];  $dT_H/dt < 0$ , asymptotic extremality at infinite time) regimes during Hawking-radiation evaporation. Left panel:  $T_H(t)$  contrast. Right panel: substrate-response coefficient  $\delta_{\text{ADW}} = (\alpha_{\text{ADW}} - 1)\Lambda_{\text{UV}}$  vanishing at  $\alpha_{\text{ADW}} = 1$ . The critical mass  $M_c$  separating regimes is given by Eq. (3). Lean witness: `T_H_acoustic_evolution_strict_decreasing + T_H_schwarzschild_strict_decreasing`.

positivity of all input fields.

*Two-regime profiles.* For  $b.M > M_c(p)$ , the textbook Hawking 1975 result [1] (in natural units) is  $T_H^{\text{Schw}}(M) = 1/(8\pi M)$ , with `T_H_schwarzschild_pos` and `T_H_schwarzschild_strict_decreasing` formalized. Combined with the standard Hawking-emission  $dM/dt < 0$ , this establishes  $dT_H/dt > 0$  in the Schwarzschild branch.

For  $b.M < M_c(p)$ , the BEC-acoustic backreaction of Balbinot *et al.* [2] applies. Their leading-order result is the linear-in- $t$  form Eq. (1) with the rational extrapolation  $t \sim 1/T^3$  giving  $T_H \rightarrow 0$  in infinite time. The Lean module’s analytic anchor for the strict-decreasing claim is the exponential alias

$$T_H^{\text{acoustic}}(t) = T_H^{(0)} \cdot \exp(-t/\tau_{\text{cool}}), \quad (4)$$

which matches the project’s Python anchor `src/wkb/backreaction.py` (line 449): note that `backreaction.py` evolves the full coupled-ODE system rather than assuming the exponential form, with Eq. (4) serving as the strictly-decreasing analytic alias for proof purposes. We prove `T_H_acoustic_evolution_pos`, `T_H_acoustic_evolution_at_zero`, and the load-bearing `T_H_acoustic_evolution_strict_decreasing`: for  $T_H^{(0)} > 0$ ,  $\tau_{\text{cool}} > 0$ , and  $t_1 < t_2$ ,  $T_H^{\text{acoustic}}(t_2) < T_H^{\text{acoustic}}(t_1)$ . This is the BEC-acoustic regime’s primary-source-grounded sign claim, with the exponential alias preserving the strictly-decreasing structure of the Balbinot 2005 linear-in- $t$  leading order plus the rational  $t \sim 1/T^3$  asymptote.

*Tracked-hypothesis bundle.* Per the project’s discipline for non-derived structural identifications, we bundle the regime partition into a Lean structure with five

mutually-independent fields (anti- $\exists$ -absorption-pattern audit clean):

$$\begin{aligned} H_{\text{RegimePartition}}(b, p, T_H^{(0)}, \dot{T}_{\text{evap}}, \delta) := & \\ & M_{\text{c,form-consistent}} : 0 < M_c(p) \\ & \wedge T_{H0\_pos} : 0 < T_H^{(0)} \\ & \wedge \text{evap\_dT\_dt\_above} : b.M > M_c(p) \rightarrow 0 \\ & \wedge \text{evap\_dT\_dt\_below} : b.M < M_c(p) \rightarrow \dot{T} \\ & \wedge \delta\_consistent\_with\_ansatz : \delta = (\alpha_{\text{ADW}} \end{aligned}$$

The bundle is genuinely non-trivial: four falsifier theorems witness four independent ways the bundle’s substantive content (the four physics-content fields; the fifth field `T_H0_pos` is a positivity assumption and is not falsifier-witnessed) can be violated. `falsifier_acoustic_decay_form` (the BEC-acoustic strictly-decreasing claim cannot be reproduced by a non-decreasing form); `falsifier_schwarzschild_heating` (a positive heating rate at the Schwarzschild branch: sign claim for  $b.M > M_c$ ); `falsifier_third_law_form` (the regime-partition bundle’s third-law slot is not vacuous-true;  $t \rightarrow \infty$  approach is required); `falsifier_alpha_ADW_dependence` (the substrate-response coefficient  $\delta$  must depend on  $\alpha_{\text{ADW}}$  via the Wave-9 ansatz, not be a free parameter). The correctness-push theorem `regime_partition_criterion` combines these with the classifier to produce the explicit sign-of- $dT_H/dt$ -flips-at- $M_c$  statement.

*Four laws by regime.* The four laws (zeroth/first/second/ third) partition across regimes as informal narrative-level statements; unlike  $H_{\text{RegimePartition}}$ , they are NOT bundled into a single Lean Prop structure  $H_{\text{BCH}}$  but appear as separate theorems and predicates in `BHThermodynamicsFourLaws.lean`. The zeroth law ( $\kappa$  constant on the horizon) and first law ( $dM = (\kappa/8\pi)dA + \Omega dJ + \Phi dQ$ ) are substrate-agnostic. The second law (entropy non-decrease) is formalized via Glorioso–Liu SK-EFT entropy-current monotonicity [8–10] on the open Schwinger–Keldysh action, without invoking pointwise NEC; this is the load-bearing substrate-class generalization compared to the standard Bekenstein–Hawking-on-area-law treatment. The third law (Israel’s strong form [5]:  $\kappa \rightarrow 0$  attainable only at infinite affine time; equivalently, no extremal-horizon formation in finite proper time) holds in both branches under the regime-specific asymptotics: the BEC-acoustic branch reaches the  $T_H \rightarrow 0$  extremality only at  $t \rightarrow \infty$  (Balbinot 2005  $t \sim 1/T^3$  extrapolation), and the Schwarzschild branch evaporates completely at finite proper time without ever crossing the  $\kappa \rightarrow 0$  locus along the evaporation trajectory. Recent third-law revisions for extremal BPS BHs [6, 7] do not bear on the Schwarzschild branch; their applicability to the ADW-extremality branch is a Phase-6 cross-bridge open

question.

## DISCUSSION

*What is novel.* Three claims are project-original and explicitly flagged: (1) the critical-mass functional form  $M_c = N_f \Lambda_{UV} / (12\pi\alpha_{ADW})$ , no published ADW derivation exists; (2) the BEC-acoustic exponential approximation Eq. (4) as the load-bearing analytic anchor for the strict-decreasing claim, an analytic compression of the coupled-ODE system in [2]; (3) the substrate-response coefficient  $\delta_{ADW} = (\alpha_{ADW} - 1)\Lambda_{UV}$  as a unified four-falsifier witness across the bundle’s degrees of freedom.

*What is not asserted.* We do *not* derive a single closed-form  $T_H(M)$  interpolating the regimes; the boundary  $M = M_c$  is a discrete-classification boundary in the inductive type, not a smooth crossover. We do *not* prove that the Glorioso–Liu second law operates strictly identically in the two regimes; entropy-current monotonicity holds in each regime under its respective hypothesis bundle, but the cross-regime entropy matching at the boundary is a Phase-6 cross-bridge.

*Comparison: Jacobson–Koike contrast.* The moving-domain-wall analog in  $^3\text{He-A}$  studied by Jacobson–Koike [4] gives a heating profile ( $T_H(v) = T_H^{(0)}(1 - v^2/c_\perp^2)$ ) monotonically decreasing in  $v$ , but the back-reaction *slows* the soliton  $dv/dt < 0$ , so  $T_H$  *increases* in time). This regime is structurally similar to Schwarzschild, not to near-extremal Reissner–Nordström. We cite [3, 4] only as the contrast case, not as the cooling anchor.

*Substrate-class context.* The  $^3\text{He-A}$  cell and the BEC-acoustic substrate are members of complementary Sakharov-criterion classes. The Volovik–Jannes formalisation in the program’s `lean/SKEFTHawking/JacobsonThermoGRDarkEnergy.lean` (Phase 6m Track C) shows that  $^3\text{He-A}$  satisfies all four Sakharov conditions (JTGR7: emergent IR Lorentz, universal coupling, induced Newton constant, and the  $\Lambda_{\text{eff}} = \Lambda_{\text{HK}}$  equilibrium identity), while the Finazzi–Liberati–Sindoni acoustic-BEC violates condition (ii) (universal coupling: depletion-sector atoms do not propagate on the BEC effective metric; JTGR8). The Phase 6n substrate partition (`lean/SKEFTHawking/CrooksAnalogHawking/SakharovHorizonCrooks.lean`) ships a 2-conjunct structural partition statement (`horizonCrooks_substrate_partition`:  $^3\text{He-A}$  is Jacobson-consistent while the FLS BEC is not) under the local Rindler-horizon horizon-Crooks framing; the companion deep paper [14] carries the full subsection including the substantive biconditional lift in `BiconditionalReformulation.lean` (`sakharov_iff_horizon_crooks`). The regime partition formalised here (sign-flip of  $dT_H/dt$ ) is *not* directly equivalent to Sakharov-criterion satisfaction: the BEC-acoustic ( $M < M_c$ ) cooling regime persists *despite*

Sakharov-criterion failure on the FLS substrate, because the regime partition is a thermodynamic-evolution claim about  $T_H(t)$  rather than a substrate-induced-gravity claim. The two classifications (Sakharov-criterion + regime partition) are independent program axes; horizon-Crooks unifies them only at the substrate-induced-gravity layer (the Jacobson 1995 equilibrium-Clausius reading [13] of entropic gravity), not at the thermodynamic-evolution layer of the regime classifier. The fluctuation-theorem reading of the unification is consistent only with the equilibrium-Clausius derivation of the Einstein equation; it does not entail Verlinde-class entropic-force gravity, which the program separately disfavours at  $\geq 5\sigma$  across eight independent observational checks (Phase 6m Track B).

*Reproducibility.* The result is reproducible in two layers. The Lean layer is `BHThermodynamicsFourLaws.lean` (20 theorems and lemmas, zero `sorry`, zero new axioms, zero `maxHeartbeats` overrides) at the project’s pinned Mathlib4 commit. The Python layer is `src/wkb/backreaction.py` (the BEC-acoustic ODE evolution at line 449) and the `src/core/visualizations.py:fig_T_H_evolution_regime_partition` visualization. All numerics quoted in the abstract and figure caption are committed-pinned.

*Outlook.* The Glorioso–Liu second law’s all-orders generalization [12] and the entropy current’s pull-back to the SK-double [11] are the natural Phase-6 cross-bridges for promoting the 20-theorem regime-partition bundle to a fully open-system second-law derivation in both branches.

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